

Lecture #0

CSE276

ARTIFICIAL INTELLIGENCE

FOUNDATIONS



Course Introduction

- **Course Code :** CSE276
- **Course Title :** Artificial Intelligence Foundations
- **L-T-P Structure :** 2 0 2
- **Credits:** 03
- **Nature of Course :** Core



Vision

To be a globally recognized school through excellence in teaching, learning and research for creating Computer Science professionals, leaders and entrepreneurs of future contributing to society and industry for sustainable growth.



Mission

M1: To provide a learning-based environment on technical concepts applied to real-life situations with measurable outcomes.

M2: To establish connections with the industry for curriculum design, and creating internship cum career opportunities.

M3: To address societal issues related to regional, national and global challenges through meaningful research.

M4: To inspire graduates for pursuing lifelong learning in professional careers.

M5: To develop leadership potential in ethically competent entrepreneurs.



Course Outcomes (COs)

CO1 :: Explain the foundations, evolution, characteristics and applications of Artificial Intelligence.

CO2 :: Apply state-space formulation and classical search techniques for solving Artificial Intelligence problems.

CO3 :: Describe knowledge representation, reasoning techniques and intelligent agent models in Artificial Intelligence.

CO4 :: Explain the evolution of Machine Learning, Deep Learning and modern Artificial Intelligence paradigms.

CO5 :: Apply prompt engineering principles and Responsible AI concepts for effective use of Generative AI and Large Language Models.

CO6 :: Describe the Artificial Intelligence development ecosystem, emerging technologies and future trends.

Program Outcomes

PO1: Apply knowledge of computing, mathematics, artificial intelligence, data science, and engineering fundamentals to solve complex computing problems.

PO2: Identify, formulate, analyze, and solve complex computing and AI-related problems using appropriate analytical techniques.

PO3: Design innovative software systems, AI-enabled applications, databases, and computing solutions that meet specified requirements.

PO4: Conduct experiments, analyze data, evaluate results, and derive meaningful conclusions for computing and AI applications.

PO5: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

Program Outcomes (Cont.)

PO7: Apply ethical principles and professional responsibilities in software engineering, cybersecurity, data handling, and AI systems.

PO8: Function effectively as an individual and as a member or leader in multidisciplinary and diverse teams.

PO9: Communicate effectively with technical and non-technical stakeholders through reports, presentations, documentation, and discussions.

PO10: Apply project management principles, business practices, resource planning, and financial considerations in technology-driven projects.

PO11: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Program Specific Outcomes

PSO1: Design, develop and deploy intelligent software systems by applying computer science fundamentals and artificial intelligence techniques including machine learning, deep learning, computer vision, natural language processing, foundation models and reinforcement learning.

PSO2: Analyze complex computational problems and develop scalable AI solutions using data analytics, optimization techniques, modern AI frameworks, responsible AI practices and industry-oriented engineering tools.

Programme Education Objectives

- **PEO1:** Graduates will establish themselves as competent computing professionals by applying principles of computer science, artificial intelligence, machine learning, and emerging technologies to develop effective solutions for real-world problems.
- **PEO2:** Graduates will demonstrate innovation and entrepreneurial mindset by designing intelligent software solutions, AI-enabled applications, and data-driven systems addressing industrial and societal challenges.
- **PEO3:** Graduates will engage in lifelong learning, professional development, and upskilling to adapt to advancements in artificial intelligence, computing technologies, and evolving industry requirements.
- **PEO4:** Graduates will demonstrate professional ethics, teamwork, leadership, and responsibility while developing sustainable and trustworthy technology solutions considering societal and environmental impact.

Need of Artificial Intelligence?

The need for AI:

- 1. Automation of Repetitive Tasks**
- 2. Data Analysis and Insights**
- 3. Enhanced Decision Making**
- 4. Improved Customer Experience**
- 5. Medical Advancements**
- 6. Security and Fraud Detection**
- 7. Smart Infrastructure and Cities**
- 8. Innovations in Education**
- 9. Agricultural Improvements**
- 10. Economic Growth and Competitiveness**

Course Overview & Importance

- **The Foundation of Artificial Intelligence:**

Introduces the fundamental concepts of Artificial Intelligence, enabling machines to perceive, reason, learn, and make intelligent decisions for solving real-world problems.

- **Core AI Skills:**

Learn AI problem formulation, search algorithms, knowledge representation, intelligent agents, modern AI paradigms, Generative AI, Large Language Models (LLMs), and Responsible AI with hands-on AI application development.

- **Industry Importance:**

Artificial Intelligence powers today's intelligent applications including virtual assistants, healthcare diagnostics, autonomous systems, recommendation engines, smart manufacturing, financial analytics, cybersecurity, robotics, and Generative AI solutions.

- **Placement Readiness:**

Builds a strong foundation for advanced AI, Machine Learning, Deep Learning, NLP, Computer Vision, and Generative AI courses while preparing students for technical interviews, AI/ML internships, hackathons, and entry-level AI Engineer roles.

Course Content

Unit	Broad Topic	Sub-Topics & Key Concepts	Practical Applications
Unit I	Foundations of Artificial Intelligence	Introduction to AI, History & Evolution, Characteristics & Types of AI, AI vs ML vs DL vs Data Science, AI Lifecycle, Applications of AI, Problem Formulation, State-Space Representation, Problem Characteristics	Building intelligent applications such as virtual assistants, smart healthcare systems, recommendation systems, autonomous navigation and fraud detection.

Course Content

Unit	Broad Topic	Sub-Topics & Key Concepts	Practical Applications
Unit II	Problem Solving & Knowledge Representation	BFS, DFS, Best First Search, Hill Climbing, A* Search, Heuristic Functions, Knowledge Representation, Semantic Networks, Frames, Production Systems, Expert Systems, Propositional Logic, Predicate Logic, Bayes' Theorem	Developing intelligent route planners, puzzle solvers, expert systems, decision support systems and knowledge-based AI applications.

Course Content

Unit	Broad Topic	Sub-Topics & Key Concepts	Practical Applications
Unit III	Intelligent Agents	Agent Architecture, PEAS Framework, Rational Agents, Reflex Agents, Model-Based Agents, Goal-Based Agents, Utility-Based Agents, Learning Agents, Autonomous Intelligent Systems, Multi-Agent Systems, Human-AI Collaboration	Designing autonomous robots, smart home systems, warehouse automation, intelligent traffic management and collaborative AI systems.

Course Content

Unit	Broad Topic	Sub-Topics & Key Concepts	Practical Applications
Unit IV	Evolution of Modern Artificial Intelligence	Symbolic AI, Statistical AI, Machine Learning, Deep Learning, Foundation Models, Multimodal AI, Agentic AI, Edge AI, Tiny AI	Developing AI solutions using machine learning, deep learning and intelligent edge devices for healthcare, vision and automation.

Course Content

Unit	Broad Topic	Sub-Topics & Key Concepts	Practical Applications
Unit V	Generative AI & Responsible AI	Generative AI, Large Language Models, Prompt Engineering, Retrieval-Augmented Generation (RAG), Hallucination, Explainable AI, Fairness, Bias, Transparency, Responsible AI, Privacy, Security, AI Governance and Ethics	Building AI assistants, chatbots, content generation systems and trustworthy AI applications with explainability and ethical decision-making.

Course Content

Unit	Broad Topic	Sub-Topics & Key Concepts	Practical Applications
Unit VI	AI Ecosystem & Future Trends	Python AI Ecosystem, TensorFlow, PyTorch, Hugging Face, Ollama, LangChain, Vector Databases, AI Development Workflow, Artificial General Intelligence (AGI), Sustainable AI	Developing end-to-end AI applications using modern frameworks, deploying LLM-based solutions and exploring next-generation AI technologies.

List of Practicals

1. Develop an Intelligent Route Planner using Classical Search Algorithms

Design an AI-based route planning system for emergency response or campus navigation by implementing Breadth First Search (BFS) and Depth First Search (DFS). Compare both algorithms in terms of path quality, explored states and execution time.

2. Design an AI-Based Puzzle Solver using Heuristic Search

Develop an intelligent solver for problems such as the 8-Puzzle or maze navigation using Hill Climbing and A* Search algorithms. Evaluate how heuristic functions improve search efficiency compared to uninformed search methods.

3. Build an Intelligent Medical Expert System

Develop a rule-based expert system to assist in the preliminary diagnosis of common diseases based on symptoms using production rules and logical reasoning. Analyze how expert systems support decision-making in healthcare.

List of Practicals

4. Develop a Smart Decision Support System using Knowledge Representation

Implement knowledge representation techniques such as semantic networks, frames, propositional logic or predicate logic to model a real-world problem like university admission, loan approval or career guidance. Demonstrate intelligent reasoning using the developed knowledge base.

5. Design an Intelligent Autonomous Agent for Smart Environments

Develop a simple intelligent agent capable of making decisions in a simulated environment such as a robotic vacuum cleaner, warehouse robot or traffic signal controller. Demonstrate different agent architectures and compare their behaviors.

6. Develop a Multi-Agent Simulation for Collaborative Problem Solving

Implement a multi-agent system where multiple intelligent agents cooperate to accomplish tasks such as warehouse management, disaster response or autonomous traffic management. Analyze communication and coordination among intelligent agents.



List of Practicals

7. Explore the Evolution of AI through a Comparative Machine Learning Study

Implement simple examples demonstrating supervised and unsupervised learning using publicly available datasets. Compare these approaches conceptually with classical AI methods to understand the evolution of intelligent systems.

8. Build an AI Assistant using a Large Language Model

Develop a domain-specific AI assistant (such as a Student Assistant, Healthcare Assistant or Travel Guide) using an open-source or publicly available Large Language Model through Ollama or Hugging Face. Design effective prompts to improve response quality.

9. Enhance AI Responses using Prompt Engineering

Experiment with different prompt engineering strategies to improve the accuracy, reasoning and creativity of responses generated by a Large Language Model. Analyze the impact of prompt design on AI-generated outputs across different application scenarios.



List of Practicals

10. Investigate Responsible AI using Explainability Techniques

Develop an AI application and evaluate its fairness, transparency and interpretability using Explainable AI concepts. Analyze prediction bias and discuss ethical implications in domains such as healthcare, education or finance.

11. Develop an AI Application using the Modern AI Development Ecosystem

Build a simple AI application using tools from the modern AI ecosystem such as TensorFlow, PyTorch, Hugging Face or Ollama. Understand the complete AI development workflow from model selection to inference without focusing on complex model training.

12. Capstone Mini Project: Artificial Intelligence for Social Good

Design and implement an AI-based solution addressing a real-world societal challenge such as crop disease advisory, waste segregation, fake news detection, smart attendance, disaster response, traffic monitoring or AI-assisted education. Integrate multiple Artificial Intelligence concepts studied throughout the course and evaluate the effectiveness of the proposed solution.

Innovative Pedagogies

- **Think–Pair–Share + AI Problem Solving:** Students analyze real-world AI problems (route planning, medical diagnosis, smart assistants) and collaboratively formulate intelligent solutions using search techniques and knowledge representation.
- **Case-Based Learning (CBL) + AI Demonstrations:** Explore intelligent agents, expert systems, Large Language Models (LLMs), and Generative AI through live demonstrations and case studies to compare different AI approaches for practical applications.
- **Challenge-Based Learning + Mini AI Project:** Design and develop AI-based solutions such as chatbots, route planners, expert systems, or AI assistants using modern AI tools (Hugging Face, Ollama, TensorFlow/PyTorch), followed by peer evaluation and presentation.

Assessment & Evaluation Framework

Academic Activity	Maximum Marks
Attendance	5
Continuous Assessment (CA)	25
Mid-Term Examination (MCQ based)	20
End Term Examination (MCQ+ Subjective)	50
Total	100

Course Assessment

CA No.	Type	Course Coverage	Assessment Description	Compulsory
CA 1	Project Assessment	Unit I -VI	Implementation-10 Marks Report Writing - 10 Marks Viva & Presentation- 10 Marks	Yes
CA 2	Test Assessment-1	Unit I- II	A MCQ based test will be conducted covering Unit-I and Unit-II of the syllabus.	Best one out of 2 shall be considered
CA 3	Test Assessment-2	Unit I- V	Subjective questions will be used to evaluate students' conceptual understanding, analytical thinking, and problem-solving skills related to Artificial Intelligence, search techniques, knowledge representation, intelligent agents, Generative AI, and Responsible AI..	

Learning Resources

Textbook:

ARTIFICIAL INTELLIGENCE: A MODERN APPROACH by Stuart Russell & Peter Norvig, Pearson Education

Reference Book:

ARTIFICIAL INTELLIGENCE by Elaine Rich, Kevin Knight & Shivashankar B. Nair, McGraw-Hill Education

Demo Kits & Other Readings:

NPTEL Artificial Intelligence: Concepts and Techniques (IISc Bangalore), Hugging Face Documentation, Ollama Documentation, TensorFlow & PyTorch Official Documentation

Virtual Lab:

Google Colab, Hugging Face Spaces, TensorFlow Playground

Placement & Career Relevance

Industry Ready Skills

- Data Preprocessing & Feature Engineering
- Classification & Regression
- Ensemble Learning & Hyperparameter Optimization
- Explainable AI (LIME & SHAP)
- Time Series Forecasting
- ML Pipelines & Model Deployment
- Python & Scikit-learn

Industry Applications

- Healthcare
- Fraud Detection
- Customer Churn
- Loan Approval
- Employee Attrition
- House Price Prediction
- Demand Forecasting

Career Opportunities

ML Engineer, AI Engineer, Data Scientist, MLOps Engineer, Business Intelligence Analyst

Higher Education

NPTEL, Google ML Engineer, Azure AI, AWS ML, M.Tech/MS AI & Data Science

Revenue Generation & Entrepreneurship Opportunities

AI Products

Loan Recommendation System | Medical Diagnosis Assistant | Sales Forecasting Dashboard | Customer Churn Platform

Freelancing

ML Model Development | Predictive Analytics | Dashboard Development | AI Automation

Startups

AI Consulting | Healthcare AI | FinTech Risk Prediction | EdTech AI

Industry Collaboration

Capstone Projects | Sponsored Projects | Technology Transfer | Consultancy

Innovation, Research & Project Opportunities

Research Areas • Explainable AI • Responsible AI • AutoML • Time Series Analytics • Healthcare AI	Publications • IEEE • Springer • Scopus Journals • Book Chapters
Innovation • AI Decision Support Systems • Explainable AI Dashboards • Business Intelligence	Competitions • Smart India Hackathon • Kaggle • Google Solution Challenge • Patent Filing

Skill Attainment

Technical Skills

- Python Programming for AI
- Problem Formulation & Search Techniques
- Knowledge Representation
- Rule-Based Expert Systems
- Machine Learning Fundamentals
- Data Preparation & Visualization
- AI Model Evaluation
- Prompt Engineering Basics
- AI Development using Python Libraries

Professional Skills

- Analytical & Critical Thinking
- Computational Problem Solving
- Decision Making
- Team Collaboration
- Technical Documentation
- Presentation & Communication Skills
- Creativity & Innovation
- Responsible AI Practices

Industry-Aligned Competencies

- Develop AI-based solutions using foundational AI techniques.
- Select appropriate AI approaches for real-world problems.
- Build intelligent systems using Python and AI libraries.
- Apply AI in healthcare, education, finance, and smart systems.
- Design ethical, explainable, and responsible AI applications.

Student Expectations & Academic Conduct

Attendance & Participation

- Maintain a minimum attendance as per university regulations.
- Actively participate in lectures, coding sessions, quizzes, and classroom discussions.
- Complete pre-class reading and engage in flipped learning activities where applicable.

Assignments & Practical Work

- Submit assignments, lab exercises, and projects within the prescribed deadlines.
- Write clean, well-documented, and reproducible Python code.
- Demonstrate your solution during practical evaluations whenever required.

Academic Integrity

- Follow ethical coding and AI practices.
- Avoid plagiarism, code copying, and unauthorized use of AI-generated submissions.
- Properly acknowledge datasets, libraries, and research references used in projects.

Professional Conduct

- Maintain discipline and respect during classroom and laboratory sessions.
- Collaborate effectively during team-based activities while ensuring individual contributions.
- Develop a growth mindset by embracing feedback and continuous improvement.

Student Guidance & Support

Learning Resources

Utilize the recommended **NPTEL course – Artificial Intelligence: Concepts and Techniques (IISc Bangalore)** and other Open Educational Resources (OERs).

Refer to prescribed textbooks, Hugging Face, TensorFlow, PyTorch, and Ollama documentation for conceptual and practical understanding.

Practice using **Google Colab**, Hugging Face Spaces, and GitHub for hands-on AI development.

Industry Certifications

Pursue certifications from **NPTEL, Google Cloud, Microsoft Learn, AWS Skill Builder, IBM SkillsBuild, Oracle University, and Coursera** to strengthen AI skills.

Innovation & Research

Participate in **Smart India Hackathon (SIH), Kaggle competitions, AI hackathons, research projects, and innovation challenges**.

Explore **Generative AI, Large Language Models (LLMs), Agentic AI, and Responsible AI** through mini projects and case studies.

Faculty Support & Feedback

Seek guidance during faculty consultation hours and LMS discussion forums.

Regular feedback will be provided on assignments, practicals, quizzes, and project work to support continuous learning.



OER Mapping (Unit I)

Unit	Broad Topic	OER Type	Title of OER	% Mapped	Source URL
Unit I	Foundations of Artificial Intelligence	Video Lectures + PDFs	Artificial Intelligence: Concepts and Techniques (IISc Bangalore) – Introduction to AI, History & Evolution, Types of AI, AI vs ML vs DL, Intelligent Systems, AI Lifecycle, Applications, Problem Formulation & State Space Representation.	80–85%	https://nptel.ac.in/courses/106108705

OER Mapping (Unit II)

Unit	Broad Topic	OER Type	Title of OER	% Mapped	Source URL
Unit II	Problem Solving & Knowledge Representation	Video Lectures + PDFs	Artificial Intelligence: Concepts and Techniques (IISc Bangalore) – Uninformed & Informed Search (BFS, DFS, Best First, Hill Climbing, A*), Heuristics, Knowledge Representation, Semantic Networks, Frames, Expert Systems, Logic & Bayes' Theorem.	80–85%	https://nptel.ac.in/courses/106108705

OER Mapping (Unit III)

Unit	Broad Topic	OER Type	Title of OER	% Mapped	Source URL
Unit III	Intelligent Agents	Video Lectures + PDFs	Artificial Intelligence: Concepts and Techniques (IISc Bangalore) – Agent Architecture, PEAS Framework, Rational, Reflex, Goal-Based, Utility-Based & Learning Agents, Autonomous Systems and Multi-Agent Systems.	80–85%	https://nptel.ac.in/courses/106108705

OER Mapping (Unit IV)

Unit	Broad Topic	OER Type	Title of OER	% Mapped	Source URL
Unit IV	Evolution of Modern Artificial Intelligence	Video Lectures + PDFs	Artificial Intelligence: Concepts and Techniques (IISc Bangalore) – Symbolic AI, Statistical AI, Machine Learning, Deep Learning, Foundation Models, Multimodal AI, Agentic AI, Edge AI and Tiny AI.	80–85%	https://nptel.ac.in/courses/106108705



OER Mapping (Unit V)

Unit	Broad Topic	OER Type	Title of OER	% Mapped	Source URL
Unit V	Generative AI & Responsible AI	Video Lectures + PDFs	Artificial Intelligence: Concepts and Techniques (IISc Bangalore) – Generative AI, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Explainable AI, Responsible AI, AI Ethics and Governance.	80–85%	https://nptel.ac.in/courses/106108705



OER Mapping (Unit VI)

Unit	Broad Topic	OER Type	Title of OER	% Mapped	Source URL
Unit VI	AI Development Ecosystem & Future Trends	Video Lectures + PDFs	Artificial Intelligence: Concepts and Techniques (IISc Bangalore) – Python AI Ecosystem, TensorFlow, PyTorch, Hugging Face, Ollama, LangChain, Vector Databases, AI Development Workflow, AGI and Sustainable AI (<i>supplemented with instructor resources where required</i>).	80–85%	https://nptel.ac.in/courses/106108705

MOOC Mapping & its benefits

MOOC Details	Description
MOOC Platform	NPTEL (SWAYAM)
Course Name	Artificial Intelligence: Concepts and Technique
Course Duration	12 Weeks
Course URL	https://nptel.ac.in/courses/106108705
Mapping with INT527 Syllabus	84% (Approx.)
Eligibility for Benefits	Students must successfully complete the NPTEL course and obtain an Elite Certificate.
Academic Benefits	Complete waiver of all Continuous Assessments (CA-1, CA-2, CA-3, etc.) and Mid-Term Examination (MTE) for students who earn the Elite Certificate. *Students will be awarded marks as per the university MOOC policy.



Education Revolution

Why EDU-Revolution!!!

- Career Readiness
- Enhance Real World Skills
- Promote Innovation and Creativity
- Foster Industry Relations
- Develop Entrepreneurial Mindset
- Increase Student Engagement and Motivation
- Promote Lifelong Learning

Academic initiatives under EDU Revolution

Revenue Generation



Projects & Hackathons



Internship Opportunities (National/International)



NPTEL / SWAYAM / MOOCs



Grade-Upgradation



10 % Attendance Benefits



Duty Leaves



Social Media Presence



Recognition of Prior Learning (RPL)



Semester Abroad



Community Service



Extra / Co-curricular Activities



EDU Revolution Benefit Mapped with CSE276



RPL PROCESS:

1. **Application:** Through UMS Portal
2. **Submission of Evidence:** Upload relevant documents such as:
Coursera, NPTEL, edX, or other recognized certifications/ Project reports/ GitHub repositories/
Internship or industry experience certificates/
Workshop.
1. **Document Verification:** By Competent Authority
2. **Competency Assessment/Examination:** Upon passing in RPL Examination, Student shall be eligible for the RPL Benefit.



Feedback

Feedback Mechanism: It is mandatory for all students to participate in academic improvement processes and fill out all types of feedback regularly throughout the semester.